

ORIGINAL RESEARCH

Characteristics and Attitudes of Wearable Device Users and Nonusers in a Large Health Care System

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BACKGROUND: Consumer wearable devices with health and wellness features are increasingly common and may enhance disease detection and management. Yet studies informing relationships between wearable device use, attitudes toward device data, and comprehensive clinical profiles are lacking.

METHODS AND RESULTS: WATCH-IT (Wearable Activity Tracking for Comprehensive Healthcare-Integrated Technology) studied adults receiving longitudinal primary or ambulatory cardiovascular care in the Mass General Brigham health care system from January 2010 to July 2021. Participants completed a 20-question electronic survey about perceptions and use of consumer wearable devices, with responses linked to electronic health records. Multivariable logistic regression was used to identify factors associated with device use. Among 214 992 individuals receiving longitudinal primary or cardiovascular care with an active electronic portal, 11 121 responded (5.2%). Most respondents (55.8%) currently used a wearable device, and most nonusers (95.3%) would use a wearable if provided at no cost. Although most users (70.2%) had not shared device data with their doctor previously, most believed it would be very (20.4%) or moderately (34.4%) important to share device-related health information with providers. In multivariable models, older age (odds ratio [OR], 0.80 per 10-year increase [95% CI, 0.77–0.82]), male sex (OR, 0.87 [95% CI, 0.80–0.95]), and heart failure (OR, 0.75 [95% CI, 0.63–0.89]) were associated with lower odds of wearable device use, whereas higher median income (OR, 1.08 per 1-quartile increase [95% CI, 1.04–1.12]) and care in a cardiovascular medicine clinic (OR, 1.17 [95% CI, 1.05–1.30]) were associated with greater odds of device use.

CONCLUSIONS: Among patients in primary and cardiovascular medicine clinics, consumer wearable device use is common, and most users perceive value in wearable health data.

Key Words: consumer wearables ■ health care innovation ■ mobile technologies

Consumer wearable technologies are increasingly common, with nearly one-third of Americans reporting use of a smart watch or fitness tracker.^{1,2} Wearable devices are often marketed to promote general health and wellness, and many commercially available products measure physiologic parameters with established clinical and prognostic significance (eg,

heart rate,³ physical activity levels^{4,5}). Selected wearable devices even contain the ability to record data previously limited to the domain of medical diagnostics, such as single-lead electrocardiograms.

Although greater availability of consumer wearable devices has potential to improve public health, there are concerns about collection of physiologic data in

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RESEARCH PERSPECTIVE

What New Question Does This Study Raise?

- Within an electronically assembled cohort of >11 000 patients in primary and cardiovascular medicine clinics with linkage to detailed health records, wearable consumer device use is common, most users perceive value in wearable health data, and most users stated they would share device data with researchers studying health outcomes.
- Older age, male sex, and heart failure were associated with lower odds of wearable consumer device use, whereas higher median zip code income and care in a cardiovascular medicine clinic were associated with greater odds of device use.

What Question Should Be Addressed Next?

- Our platform may facilitate rigorous study of the relationships between wearable consumer technology and resource use, clinical outcomes, and health disparities.

Nonstandard Abbreviations and Acronyms

C3PO	Community Care Cohort Project
EWOC	Enterprise Warehouse of Cardiology
MGB	Mass General Brigham
WATCH-IT	Wearable Activity Tracking for Comprehensive Healthcare-Integrated Technology

individuals unlikely to have disease (ie, young and healthy), as well as potential exacerbation of existing health care disparities on the basis of factors such as sociodemographic status or presence of disease. Although large population-based studies have provided insight into socioeconomic and demographic patterns of consumer wearable device use,^{2,6–10} these survey-based platforms have lacked the ability to integrate comprehensive individual-level clinical data.^{2,6–8} Large pragmatic clinical studies assessing the validity of wearable technology, such as the Apple Heart Study¹¹ and Fitbit Heart Study,¹² have similar limitations. Even smaller investigations of consumer wearable device use in relation to disease-specific outcomes, including peripheral arterial disease,¹³ obesity,^{14,15} chronic obstructive pulmonary disease,¹⁶ insomnia,¹⁷ and stroke,¹⁸ typically rely on patient-reported health indices rather

than primary data from the electronic health record (EHR).

Thus, a foundational step in understanding the relationship between wearable data and comprehensive clinical outcomes involves the creation and analysis of a sizeable cohort of wearable device users in real-world care with linkage of device-collected data to information contained within the EHR. The WATCH-IT (Wearable Activity Tracking for Comprehensive Healthcare-Integrated Technology) study sought to identify ambulatory patients within a single large health care network who use, or have an interest in using, wearable devices. We aimed to understand characteristics of wearable device users as well as patterns and drivers of device use, including potential barriers to use. WATCH-IT was additionally designed to assess the feasibility of assembling a cohort of wearable devices users for longitudinal study of digitally acquired health data and comprehensive health outcomes.

Here, we report the clinical characteristics of the WATCH-IT cohort, including demographic and clinical factors associated with wearable device use, and the results of a survey assessing patient attitudes toward health-related applications of wearable devices.

METHODS

Data and Code Availability

Mass General Brigham (MGB) data contain protected health information and will not be shared publicly. Data processing scripts used to perform the analyses described herein are available at www.github.com/shaan_khurshid/watch_it_public.

Study Design and Patient Cohort

In WATCH-IT, we electronically assembled a cohort of patients to enable study of wearable device users, including wearable device data and health outcomes. We derived the candidate list of study participants using the methods previously described to construct (1) the Community Care Cohort Project (C3PO),¹⁹ a cohort comprising >500 000 adults aged ≥18 years receiving longitudinal primary care at 1 of 11 hospitals within the MGB network, and (2) the Enterprise Warehouse of Cardiology (EWOC), an analogous cohort comprising individuals receiving longitudinal cardiovascular care.²⁰ Briefly, individuals were included if they had at least 1 pair of qualifying visits (ie, primary care or cardiology, respectively) occurring between 1 and 3 years apart. Individuals in C3PO and EWOC have clinical EHR data available within a centralized warehouse. To select for patients most likely to be active within the MGB system, candidate samples were filtered to identify patients who were alive and had started clinical follow-up (ie, the date of the second primary care visit of the earliest

qualifying pair) after January 1, 2010.¹⁹ The resulting list of ~300 000 patients was uploaded into a custom tool developed for Epic (Verona, WI), the clinical EHR platform used by MGB. Patients were matched to the complete MGB Epic database using dual identifiers including medical record number and date of birth. Once uploaded into the EHR template, active users of the MGB electronic patient portal were identified, facilitating direct-to-patient messaging via the Epic platform.

After identifying registered users of the electronic patient portal and excluding individuals who had previously opted out of research communications, we used the portal messaging system to send electronic invitations to participate in the WATCH-IT survey (date range of invitations November 2021–April 2022). The initial invitation letter included a direct link to the WATCH-IT survey instrument (Figure S1). Survey nonresponders were sent a reminder letter up to 2 weeks after initial contact date. Upon completion of the survey, participants were entered into a lottery for an Amazon gift card in the amount of \$25. Survey respondents provided informed consent, and all study protocols were approved by the MGB Institutional Review Board. This study adheres to the Strengthening the Reporting of Observational Studies in Epidemiology reporting guidelines.²¹

Survey Instrument

The survey instrument was built and managed using Research Electronic Data Capture, a secure, web-based software platform designed to support data capture for research studies.^{22,23} The survey used branching logic with initial bifurcation based on active use of a wearable device, defined as a yes response to the survey question, “Do you currently use a wearable device?” (Figure 1 and Figure S1). Individuals who opened a survey but did not answer this initial question were classified as nonresponders. Users were then asked a series of questions on patterns of current device use, whereas nonusers (defined as a no response to the question above) were asked about access to wearable device technology and reasons for nonuse. We defined a subcategory of nonusers, called potential users, who reported they would use a wearable device if one were provided at no cost. Users and potential users were asked about potential medical applications of device use, and all respondents were asked a series of general health-related questions.

Survey respondents self-identified by name and date of birth, which were used to link responses to complete health information contained in the EHR. Surveys that could not be linked by this mechanism were excluded from analysis (Figure 1).

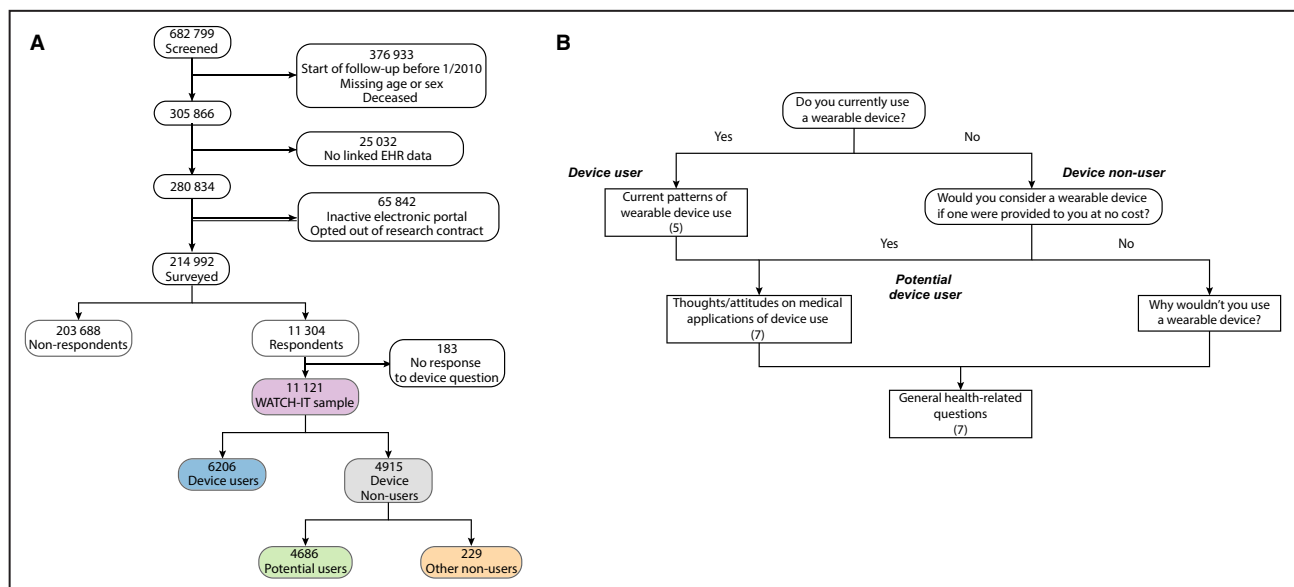


Figure 1. Study summary.

Depicted is an overview of sample construction (A) and the survey instrument (B). Using the EHR, we screened a total of 682 799 individuals with at least 2 visits within 1 to 3 years in a cardiology or primary care practice. Of these, a total of 214 992 were contacted and received a survey instrument. Of the individuals contacted, 11 121 responded to at least the initial question, comprising the WATCH-IT sample. Of these, individuals were then classified as device users vs nonusers, as described in (B). Samples comprising the main analysis sets are highlighted in color. The short WATCH-IT survey comprised branching questions (rounded rectangles) and nonbranching questions (rectangles). Responses to branching questions served to classify individuals as device users, device nonusers, and potential device users, and impacted which questions were administered. EHR indicates electronic health record; and WATCH-IT, Wearable Activity Tracking for Comprehensive Healthcare-Integrated Technology.

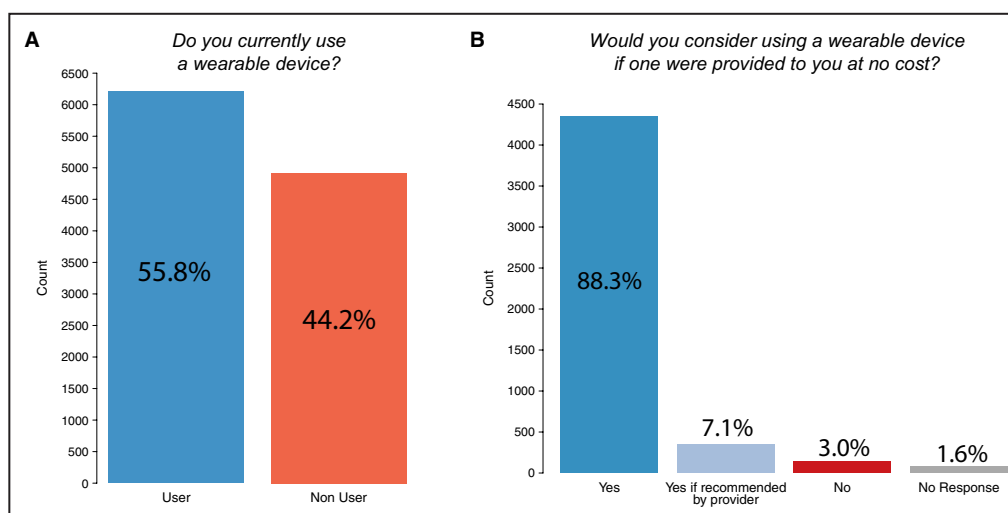


Figure 2. Frequency of active and potential wearable device use.

Depicted is the frequency of actual and potential wearable use in the analysis sample. **A**, The distribution of responses to the question, “Do you currently use a wearable device?” **B**, The distribution of responses to the question, “Would you consider using a wearable device if one were provided to you at no cost?”

Clinical Factor Ascertainment

Demographic and clinical characteristics were derived from the EHR. Demographic data included age, sex, race, and ethnicity, and were extracted from demographic fields within the patient chart. Race was classified as Asian, Black, White, unknown race (no race reported), or other race (inclusive of American Indian or Alaska Native, Native Hawaiian or other Pacific Islander, individuals reporting 2 or more race categories, or individuals reporting a race other than the previously mentioned categories). Clinic membership (primary care versus cardiology) was assigned according to whether individuals were first identified from the primary care versus cardiovascular parent samples. Median income was assigned based on zip code according to 5-year US Census data from 2017 to 2021 (data.census.gov, accessed November 2, 2022), and then divided into quartiles. Median imputation was performed for 131 individuals (1.1%) in the survey sample and 2666 (1.3%) of the nonresponder sample without income data; otherwise, there was no missingness. Prestudy follow-up was defined as the number of years between the start of cohort follow-up (ie, the second of the qualifying pair of office visits¹⁹) and the WATCH-IT study start date. Patient portal contact was quantified as the number of unique interactions with patient messages or results within the preceding year, which was then categorized into approximate quartiles (ie, 0, 1–4, 5–19, ≥ 20). Prevalent diseases were defined based on composite data comprising *International Classification of Diseases, Ninth and Tenth Revision* (ICD-9 and ICD-10) diagnosis codes, *Current Procedural Terminology* codes, and medications. Codes used to define atrial fibrillation, coronary disease, diabetes, hyperlipidemia, hypertension, heart failure, peripheral arterial disease, and

stroke have been previously validated (positive predictive value $\geq 85\%$).^{19,24,25} All disease definitions are provided in Table S1.

Statistical Analysis

The distribution of survey responses was plotted using bar graphs. Question-specific nonresponse rates $\geq 0.5\%$ were plotted as separate bars. Associations between clinical factors and wearable device use were assessed using multivariable logistic regression, with device use as the outcome and age, sex, race, income, clinic membership (primary care versus cardiovascular medicine), prestudy follow-up, portal contact, and presence of atrial fibrillation, coronary disease, diabetes, hyperlipidemia, hypertension, heart failure, other arrhythmias, obesity, peripheral artery disease, pulmonary disease, and stroke as covariates. Clinical covariates were selected a priori based on hypothesized potential associations with device use. Ethnicity was not included in the models given a low frequency of Hispanic individuals ($<1\%$). We then fit analogous models with device use (versus potential device use), potential device use (versus no potential device use), and survey response (versus nonresponse) as alternative outcomes including the same adjustment variables.

We considered 2-sided P values <0.05 to indicate statistical significance. All analyses were performed using R version 4.0 (packages `data.table`, `plyr`, and `stringr`).

RESULTS

A total of 280834 individuals met the criteria for longitudinal primary or ambulatory cardiovascular care. Of

these, 65 842 patients were excluded for either inactive electronic portal or having opted out of research contact, resulting in 214 992 individuals who were sent a survey invitation. Of invited patients, 11 121 at least partially completed a survey (5.2% response rate). Survey respondents comprised the WATCH-IT sample, with a mean age 57.6 years (SD, 15.4), 62.9% women, and 92.5% White. Total available follow-up for WATCH-IT participants was a median of 4.5 years (quartile 1: 1.9, quartile 3: 8.6). Complete baseline characteristics of the WATCH-IT cohort are provided in the Table. Characteristics of survey responders versus nonresponders are provided in Table S2.

Of the 11 121 individuals in the WATCH-IT sample, most (83.1%) reported prior use of telehealth visits and/or use of an online portal for communication with health care providers. More than half of survey respondents (n=6206, 55.8%) reported current use of a wearable device (Figure 2). Among device nonusers (n=4915),

almost all (n=4686, 95.3%) would use a wearable device if one were provided at no cost (classified as potential users). Among the 229 nonusers who would not use a device even if provided at no cost, several reasons were cited for nonuse including lack of interest (34.9%), discomfort (29.55), and other reasons (39.0%) (Figure S2). Only 7.8% of all respondents reported that a doctor had previously recommended use of a wearable device for medical reasons (Figures S3 and S4).

Of the 6206 device users, most (72.0%) stated they wore their device for at least 12 hours per day in the preceding 7 days. Most reported use of either an Apple (56.2%) or Fitbit (28.2%) brand device (Figure S5). Whereas only 24.8% of device users had previously shared device data with their doctors, more than half (54.8%) felt it was important for health care providers to be aware of information collected by their device (Figure S6). Users identified the following categories as particularly important to share with

Table 1. Baseline Characteristics of the Primary Analysis Sample

Characteristic	Survey respondents		
	Device users (n=6206)	Device nonusers (n=4915)	Overall (n=11 121)
Age, y, mean±SD	55.3±15.5	60.4±14.8	57.6±15.4
Women, n (%)	4078 (65.7%)	2918 (59.4%)	6996 (62.9%)
Race, n (%)*			
Asian	181 (2.9%)	155 (3.1%)	336 (3.0%)
Black	193 (2.9%)	149 (3.0%)	342 (3.0%)
Other	78 (1.3%)	45 (0.9%)	123 (1.1%)
Unknown	17 (0.3%)	20 (0.4%)	37 (0.3%)
White	5737 (92.4%)	4546 (92.5%)	10 283 (92.5%)
Ethnicity, n (%)			
Hispanic	53 (0.9%)	29 (0.6%)	82 (0.7%)
Non-Hispanic	5395 (86.9%)	4306 (87.6%)	9701 (87.2%)
Unknown	758 (12.2%)	758 (15.4%)	1338 (12.0%)
Median zip code income, \$ (quartile 1, quartile 3)	104 209 (84 354, 127 646)	102 088 (81 027, 124 528)	102 672 (83 469, 126 185)
Cardiology clinic, n (%)	2242 (36.1%)	1962 (39.9%)	4204 (37.8%)
Median prestudy follow-up, y (quartile 1, quartile 3)	4.9 (2.4, 9.1)	5.3 (2.7, 9.6)	5.1 (2.5, 9.3)
Median patient portal contacts (quartile 1, quartile 3) [†]	4 (0, 18)	5 (0, 19)	4 (0, 18)
Obesity, n (%)	1816 (29.3%)	1472 (29.9%)	3288 (29.6%)
Hypertension, n (%)	2973 (47.9%)	2845 (57.9%)	5818 (52.3%)
Diabetes, n (%)	1068 (17.2%)	1045 (21.3%)	2113 (19.0%)
Hyperlipidemia, n (%)	3391 (54.6%)	3165 (64.4%)	6556 (59.0%)
Coronary disease, n (%)	1624 (26.2%)	1595 (32.5%)	3219 (28.9%)
Heart failure, n (%)	320 (0.5%)	380 (0.8%)	700 (6.3%)
Pulmonary disease, n (%)	1998 (32.2%)	1686 (34.3%)	3684 (33.4%)
Atrial fibrillation, n (%)	1003 (16.2%)	883 (18.0%)	1886 (17.0%)
Other arrhythmia, n (%)	1941 (31.3%)	1681 (34.2%)	3622 (32.6%)
Stroke, n (%)	519 (8.4%)	549 (11.2%)	1068 (9.6%)

*Reflects self-reported race. Category Other race includes American Indian or Alaska Native, Native Hawaiian or other Pacific Islander, or individuals reporting ≥2 race categories, and individuals reporting a race not included among the listed categories.

[†]Number of interactions with results or messages using patient portal in the year preceding the WATCH-IT (Wearable Activity Tracking for Comprehensive Healthcare-Integrated Technology) study period.

providers: heart rate (selected by 89.7% of users), activity levels (64.4%), health conditions (53.6%), pulse oximetry (60.0%), and sleep (49.3%) (Figure S6). Most users (93.7%) agreed that they would get peace of mind knowing their device could detect a heart problem if they had one, and only a small minority (8.8%) reported anxiety related to potential detection of unknown conditions (Figure S7). After receipt of an abnormal notification from their device, 87.7% of users would contact their medical provider as a next step (Figure S8). Almost all users (91.2%) would be willing to share information from their device with researchers studying health outcomes (Figure S9). Responses were generally consistent for potential users (Figures S10 through S12).

In multivariable models evaluating factors predictive of wearable device use (versus nonuse), greater median zip code income (odds ratio [OR], 1.08 per 1-quartile increase [95% CI, 1.05–1.12]), history of atrial fibrillation (OR, 1.23 [95% CI, 1.08–1.39]), and receipt of cardiovascular care (OR, 1.16 [95% CI, 1.04–1.29]) were independently associated with greater odds of device use; whereas older age (OR, 0.80 per 10-year increase [95% CI, 0.77–0.82]), male sex (OR, 0.86 [95%

CI, 0.79–0.93]), longer prestudy follow-up (OR, 0.95 per 5-year increase [95% CI, 0.91–0.95]), and heart failure (OR, 0.75 [95% CI, 0.63–0.89]) were associated with lower odds of device use (Figure 3). Similar associations were observed when comparing device users to potential users (Figure 3). Among nonusers, older age, male sex, longer prestudy follow-up, and White race were associated with lower odds of being a potential consumer wearable device user, whereas obesity, hyperlipidemia, pulmonary disease, and cardiology patient status were associated with greater odds of being a potential user (Figure 3). When assessing factors associated with WATCH-IT survey response, older age, male sex, longer prestudy follow-up, higher income, hypertension, heart failure, peripheral arterial disease, and stroke were associated with lower odds of survey response, whereas White race, cardiology patient status, greater patient portal contact, atrial fibrillation, hyperlipidemia, other arrhythmias, and obesity were associated with greater odds of survey response (Figure 3). When compared with the primary analysis sample, individuals with heart failure were older, more likely to be men, and had a substantially greater burden of comorbidity (Table S3).

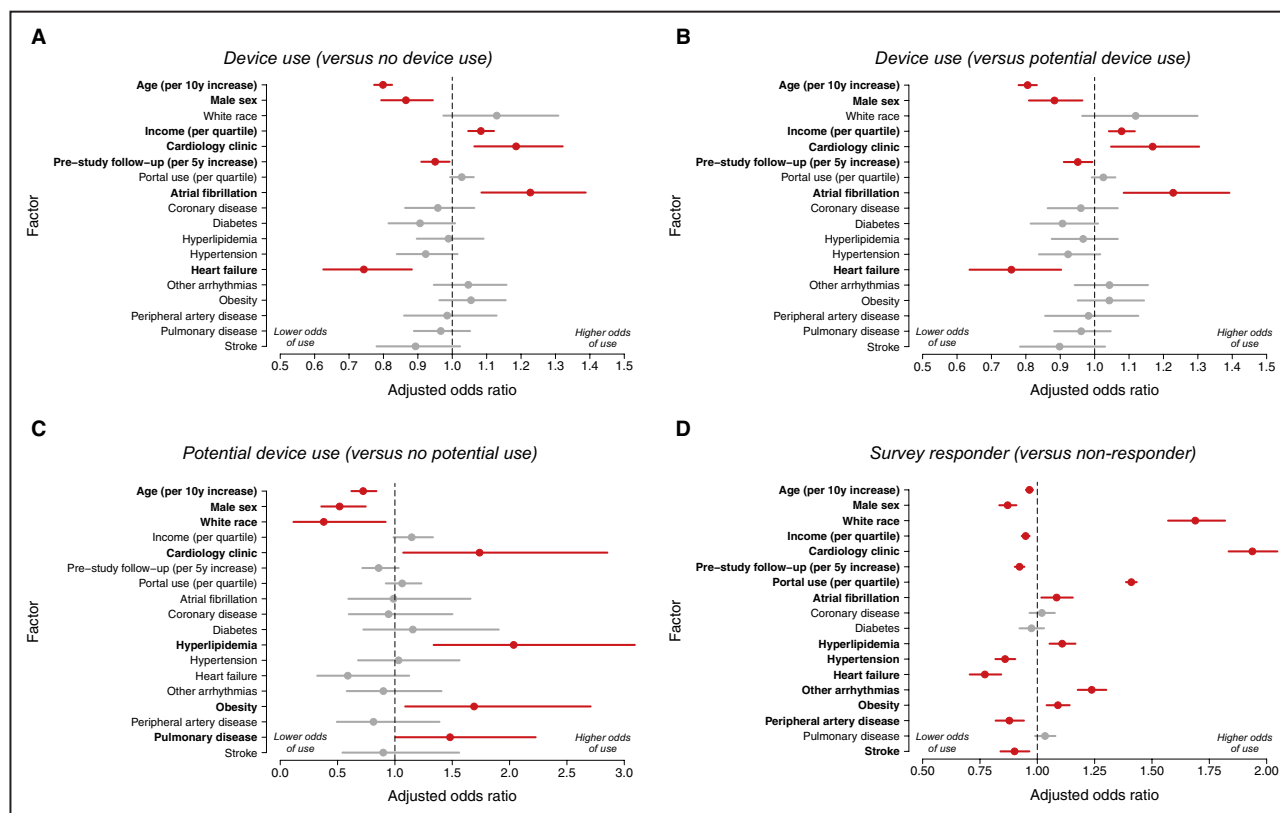


Figure 3. Factors associated with device use, potential device use, and survey response. Plotted are results of multivariable logistic regression with the following outcomes: wearable device use vs nonuse (A), wearable device use vs potential device use (B), potential device use vs no potential device use (C), and survey response vs nonresponse (D). Each model is adjusted for each of the listed factors on the y axis as covariates. Factors with significant associations are bolded and plotted with red points (vs gray points). Error bars indicate 95% CIs.

DISCUSSION

In the current study, we leveraged direct-to-patient communication at scale to develop a novel cohort of >11 000 primary care patients and patients with cardiovascular disease across a large multi-institutional health care system with linked EHR data. We found that roughly half of participants reported active use of a consumer wearable device. Atrial fibrillation and cardiology patient status were associated with greater odds of wearable device use; whereas older age, male sex, lower median zip code income, and heart failure were associated with lower odds of wearable device use.

Our study supports and extends prior work characterizing patterns and attitudes related to health technology and wearable device use. A recent analysis of ~2000 individuals from the Framingham research cohort found that age, sex, and educational attainment were each associated with digital health engagement.⁶ A second survey-based study including just >1200 respondents reported that patients with atrial fibrillation were more likely to share wearable device data with their health care providers.⁷ A nationally representative survey relying solely on self-reported data found that an estimated 29% of individuals use wearable devices, and that those with cardiovascular conditions had a lower prevalence of device use.² In our comparably large sample of individuals with linked EHR data, we found that device use is common, and that demographic and clinical factors such as age, sex, income, clinic membership, and cardiac comorbidities such as atrial fibrillation and heart failure substantially impacted the odds of wearable device use.

Our results have several implications. First, our findings suggest a discordance between the high health-related value of wearable technology perceived by patients, and their relatively limited use and uncertain clinical usefulness in current clinical practice. Specifically, survey respondents felt that device-based monitoring of physiologic parameters such as heart rate and pulse oximetry offered peace of mind and provided important information to share with health care providers. Yet only a minority of wearable device users had previously shared device data with their doctors. Additionally, although few survey respondents reported that their doctors had recommended wearable device monitoring, most device users would contact their health care provider as a next step after receiving an abnormal device alert.

Potential barriers to the implementation of wearable devices as health tools may include absence of rigorous evidence of therapeutic benefit, patient or provider unfamiliarity with specific devices and their capabilities, and a relative lack of consensus guidelines defining appropriate use of wearable devices in a clinical setting.²⁶ Of note, although the ability to collect a

wide range of biophysical data for longer durations and across broader patient populations offers a potential advantage of consumer-based wearable devices over traditional forms of ambulatory monitoring (eg, patch monitors, blood pressure recording devices, daily weights), provider bandwidth and lack of accepted reimbursement models for interpreting such data may serve as key barriers. There are also important uncertainties about potential provider liability for acting on wearable-based data and a general absence of ready infrastructure for integrating wearable-based data into existing EHR platforms and clinical workflows.^{27,28}

To this end, future prospective studies powered to demonstrate clinical benefit and evaluate strategies for integrating wearable device data into clinical practice will be critical to ensure that the patient-perceived value of wearable devices is supported by robust evidence and commensurate with their true usefulness.

Second, our analyses suggest that careful attention to specific demographic, socioeconomic, and clinical factors influencing wearable device use will be critical to ensure equitable access and distribution of care. The potential for clinical use of wearable devices to worsen existing health disparities is supported by our finding that >95% of survey respondents identified as nonusers reported that they would use a wearable device if it were offered at no cost. Additionally, in models adjusted for clinical variables, the odds of device use were substantially higher among individuals living in zip codes with greater income, whereas patient-specific factors previously associated with disparate access to care and poor health outcomes, including older age and the presence of heart failure, were associated with lower odds of device use. Our models suggest that similar factors were associated with survey response itself, suggesting the importance of sociodemographic factors and comorbidities may be even greater in the general population. In the context of prior work reporting lower odds of device use among individuals with cardiovascular disease,² our findings suggest a more complex relationship, in which factors such as engagement in longitudinal cardiovascular care or the presence of specific cardiovascular conditions (eg, atrial fibrillation) may actually increase odds of wearable use. Additional work is warranted to better understand potential barriers to access in vulnerable patient populations and to ensure that future initiatives leveraging mobile technology to improve health outcomes are implemented equitably.

Third, we submit that the WATCH-IT study provides an important proof-of-concept for electronic contact deployed at scale to assemble a cohort of wearable device users with linked EHR data for outcomes research. There is increasing recognition of the potential value of integrating wearable data with clinical outcomes, with research initiatives such as the National

Institutes of Health-based All of Us program encouraging participants to upload wearable device data as well as EHR-based health information to enable outcomes assessment.^{29,30} Notably, because EHR data may come from a multitude of sources and are not intrinsically linked, harmonization of various data types (eg, diagnoses, laboratory reports) is a substantial challenge.³¹ To this end, WATCH-IT complements prior approaches by comprising a large sample of device users already engaged in longitudinal ambulatory care, through which robust clinical EHR data are regularly ascertained. Importantly, we observed that nearly all respondents reported willingness to share device-based information with researchers studying health outcomes, which is substantially higher than prior consumer-based surveys.³² We suspect that differences may relate to varying patient characteristics or the presence of longitudinal relationships with providers and the health care system in the WATCH-IT sample.

Our study should be interpreted in the context of design. First, our observations are limited to portal users and survey respondents. Although a 5% response rate is low, it is expected for remote surveys,³³ and inherent bias exists with surveys. Future approaches leveraging hybrid in-person and electronic contact, greater incentives, or dedicated efforts focused on subgroups with lower response rates may increase survey engagement. Second, although we used previously validated algorithms to define clinical factors, some degree of misclassification of exposures remains likely. Third, most individuals included were White, and there was minimal representation of Hispanic individuals. Although the absolute number of individuals in our study from racial minority groups compares favorably to prior studies of wearable technology,^{6,7} our findings may not generalize to populations with varying racial and ethnic composition. Fourth, we studied a sample of longitudinal primary care and cardiology patients having a specific comorbidity profile that may differ from other populations. Although prior analyses suggest our sample includes less bias and greater validity as compared with other her-based sampling approaches,¹⁹ generalizability of our findings to other populations may be affected. Fifth, we did not collect data on whether participants received assistance from caregivers or family members with survey completion, which may be a relevant factor affecting wearable use or survey response.

In summary, the WATCH-IT study used a scalable platform of EHR-based communication to enroll >11 000 patients receiving longitudinal primary or cardiovascular care, finding that use of consumer-based wearable devices is common within this cohort across a large health care system. Interest in clinical usage of wearable devices was nearly universal, and attitudes

toward sharing device-related data were positive. Future research leveraging samples like the WATCH-IT cohort could empower rigorous investigation of associations between wearable device use and longitudinal clinical outcomes and provide a platform to evaluate mobile-health-based interventions.

ARTICLE INFORMATION

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Supplemental Material

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